

cron Cheat Sheet

Run commands automatically on a schedule: every few minutes, nightly, weekly, at boot.

Using it

Command	Does	Command	Does
crontab -e	Edits your schedule in an editor	crontab -l	Lists your schedule
sudo crontab -e	Edits root's schedule	crontab -u bob -l	Lists bob's schedule (as root)
crontab -r	Deletes it all, no undo! (r is next to e)	crontab -l > cron.bak	Backs up your schedule

The five time fields

```
+----- minute      0-59
|  +----- hour      0-23
|  |  +----- day of month  1-31
|  |  |  +----- month      1-12 (or jan-dec)
|  |  |  |  +---- day of week  0-7 (0 and 7 = Sunday, or sun-sat)
|  |  |  |  |
30 2  *  *  1-5 /usr/local/bin/backup.sh ← 2:30 AM, Monday to Friday
```

Symbol	Means	Symbol	Means
*	Every value (every minute, every hour...)	-	A range: 1-5 = Mon to Fri
,	A list: 0,30 = at :00 and :30	/	A step: */15 = every 15

Examples

Schedule	Runs
/ * * * *	Every 5 minutes
0 * * * *	Every hour, on the hour
0 3 * * *	Every day at 3:00 AM
0 9-17 * * 1-5	On the hour, 9 AM to 5 PM, weekdays
0 0 1 * *	Midnight on the 1st of every month
30 4 * * 0	Sundays at 4:30 AM

Shortcut	Same as	Shortcut	Same as
@reboot	Once, when the machine starts	@weekly	0 0 * * 0 (Sunday midnight)
@hourly	0 * * * *	@monthly	0 0 1 * *
@daily	0 0 * * * (midnight)	@yearly	0 0 1 1 *

Why didn't my job run? (environment gotchas)

Problem	Fix
cron's PATH is just /usr/bin:/bin	Use full paths, or set PATH=... at the top of the crontab
% means "new line" in a crontab line	Escape it: date +\%F
Output is emailed to you (or lost)	Redirect it: >> /var/log/job.log 2>&1
Your .bashrc and aliases aren't loaded	Put the work in a script and call that script
/etc/crontab and crond add a user field	0 3 * * * root /usr/local/bin/backup.sh

Checking it ran

Command	Does
journalctl -u cron --since today	cron's log on Debian/Ubuntu (-u crond on RHEL/Fedora)
grep CRON /var/log/syslog	Same, on systems with a syslog file (/var/log/cron on RHEL)
systemctl status cron	Is the cron service running? (crond on RHEL/Fedora)

```
Mar 04 02:30:01 web1 CRON[4121]: (root) CMD (/usr/local/bin/backup.sh) ← (user) and the command run
Mar 04 02:30:02 web1 CRON[4120]: (CRON) info (No MTA installed, discarding output) ← output lost
```

Remember: test by hand, use full paths, log the output. Most broken jobs are a PATH problem.